

Differentiation - Set 2 - Solutions

Specification MM6.1-MM6.3, MM8.5.

Answer key

Question	Answer	Topic
1	C	Differentiation
2	E	Differentiation
3	A	Differentiation
4	E	Differentiation
5	C	Differentiation
6	E	Differentiation
7	B	Differentiation
8	G	Differentiation
9	C	Differentiation
10	F	Differentiation

Worked solutions

Question 1 DIF-18 · Differentiation · spec MM6.3

Find the minimum value of $x + \frac{4}{x}$ for $x > 0$.

Answer: C

Derivative $1 - \frac{4}{x^2} = 0$ at $x = 2$; value $2 + 2 = 4$ (second derivative $\frac{8}{x^3} > 0$, so a minimum).

Full method - how and why

Minimising $x + \frac{4}{x}$. Differentiate, solve, and confirm it is a minimum.

- 1 Write as powers: $y = x + 4x^{-1}$ and differentiate: $y' = 1 - 4x^{-2}$.

why: $\frac{d}{dx}(4x^{-1}) = -4x^{-2}$.

- 2 Solve $y' = 0$: $1 = \frac{4}{x^2} \Rightarrow x^2 = 4 \Rightarrow x = 2$ (taking the positive root since $x > 0$).

why: The restriction $x > 0$ discards $x = -2$.

- 3 Confirm a minimum: $y'' = 8x^{-3} > 0$ for $x > 0$.

why: Positive second derivative means the curve bends upwards.

- 4 Evaluate: $y = 2 + \frac{4}{2} = 4$.

why: (Also follows from AM-GM: $x + \frac{4}{x} \geq 2\sqrt{4} = 4$.)

Where it goes wrong

- Giving the minimising $x = 2$ instead of the minimum value 4 - both are offered.
- Including $x = -2$, where the expression equals -4 but $x > 0$ is required.

Takeaway $x + \frac{k}{x} \geq 2\sqrt{k}$ for $x > 0$, with equality at $x = \sqrt{k}$ - worth recognising instantly.

Question 2 DIF-20 · Differentiation · spec MM6.3

Find the sum of the y-coordinates of the stationary points of the curve $y = 2x^3 - 9x^2 + 12x - 3$.

Answer: E

$y' = 6x^2 - 18x + 12 = 6(x-1)(x-2)$: stationary points at $x = 1$ ($y = 2 - 9 + 12 - 3 = 2$) and $x = 2$ ($y = 16 - 36 + 24 - 3 = 1$). Sum 3.

Full method - how and why

Find both stationary points and add their y-values - two substitutions after one factorisation.

- 1 Differentiate: $y' = 6x^2 - 18x + 12 = 6(x^2 - 3x + 2) = 6(x-1)(x-2)$.

why: Take out the 6, then factorise.

- 2 Stationary points at $x = 1$ and $x = 2$.

why: Roots of the derivative.

- 3 Compute the y-values: $y(1) = 2 - 9 + 12 - 3 = 2$; $y(2) = 16 - 36 + 24 - 3 = 1$.

why: Substitute into the original cubic, carefully: $2(8) = 16$ and $9(4) = 36$.

- 4 Add: $2 + 1 = 3$.

why: The question asks for the sum of the y-coordinates.

Where it goes wrong

- Adding the x-coordinates ($1 + 2 = 3$) - the same answer here by coincidence, but the wrong method.
- Arithmetic slips in $y(2)$; recompute $16 - 36 = -20$, $-20 + 24 = 4$, $4 - 3 = 1$.

Takeaway Two stationary points mean two substitutions into the original function - the derivative gives only the positions.

Question 3 DIF-08 · Differentiation · spec MM6.2

Given $f(x) = x^{\frac{3}{2}} - 6x^{\frac{1}{2}}$ for $x > 0$, find the value of x at which f has its minimum.

Answer: A

$$f'(x) = \frac{3}{2}x^{1/2} - 3x^{-1/2} = \frac{3}{2\sqrt{x}}(x - 2) = 0 \text{ at } x = 2; f' \text{ changes from negative to positive there, so it is a minimum.}$$

Full method - how and why

Differentiating fractional powers. Factorising the derivative makes the sign change easy to see.

- 1 Differentiate: $f'(x) = \frac{3}{2}x^{1/2} - 3 \times \frac{1}{2}x^{-1/2} = \frac{3}{2}x^{1/2} - 3x^{-1/2}$.
why: $\frac{d}{dx}x^{3/2} = \frac{3}{2}x^{1/2}$ and $\frac{d}{dx}(6x^{1/2}) = 3x^{-1/2}$.
- 2 Factor out $\frac{3}{2\sqrt{x}}$: $f'(x) = \frac{3}{2\sqrt{x}}(x - 2)$.
why: $\frac{3}{2}x^{1/2} = \frac{3}{2\sqrt{x}} \cdot x$ and $3x^{2\sqrt{x}} = \frac{3}{2\sqrt{x}} \cdot 2$. Factorising isolates the sign-changing part.
- 3 Set $f' = 0$: since $\frac{3}{2\sqrt{x}} > 0$ for $x > 0$, the only root is $x = 2$.
why: A positive factor can never make the product zero.
- 4 Check it is a minimum: f' is negative for $x < 2$ and positive for $x > 2$, so the curve falls then rises.
why: The sign of f' is the sign of $(x - 2)$ - immediate from the factorised form.

Where it goes wrong

- Making index errors: $\frac{d}{dx}x^{1/2} = \frac{1}{2}x^{-1/2}$, not $\frac{1}{2}x^{1/2}$.
- Squaring both sides carelessly when solving $\frac{3}{2}\sqrt{x} = \frac{3}{\sqrt{x}}$ and getting $x = 4$.

Takeaway Factor a common power out of the derivative: the remaining bracket controls the sign.

Question 4 DIF-11 · Differentiation · spec MM6.3

Which of the following statements about $f(x) = x^3 - 3x$ is/are true?

- I f is increasing for all $x > 1$.
- II f has a local maximum at $x = -1$.
- III $f(x) \geq -2$ for all real x .

Answer: E

$f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1)$: positive for $x > 1$ (I true); $f'' = 6x < 0$ at $x = -1$ (II true). III false: the local minimum $f(1) = -2$ is not a global minimum since $f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$ (e.g. $f(-3) = -18$).

Full method - how and why

Three claims about a cubic. The derivative settles the first two; the third is about global behaviour, where cubics are unbounded.

- 1 Differentiate: $f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1)$.
why: The roots ± 1 are the stationary points.
- 2 I: for $x > 1$ both factors are positive, so $f' > 0$ and f is increasing. **True**.
why: Sign analysis of the factorised derivative.
- 3 II: $f'' = 6x$, so $f''(-1) = -6 < 0$: local maximum at $x = -1$. **True**.
why: Negative second derivative means a local maximum.
- 4 III: the local minimum is $f(1) = -2$, but as $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$ - for instance $f(-3) = -27 + 9 = -18$. **False**.
why: A **local** minimum is not a global one; odd-degree polynomials are unbounded below.
- 5 I and II only.
why: Match to the options.

Where it goes wrong

- Assuming the local minimum value bounds the whole function.
- Confusing the maximum and minimum positions - for $+x^3$ the maximum is at the smaller x .

Takeaway Cubics have no global maximum or minimum; 'local' claims need f' , 'global' claims need the end behaviour.

Question 5 DIF-13 · Differentiation · spec MM6.1

$f(x)$ is a polynomial with $f'(x) = (x - 2)^2(x + 1)$. Which of the following is true?

Answer: C

f' changes sign from negative to positive at $x = -1$ (local minimum). At $x = 2$ the factor $(x - 2)^2$ does not change sign, so f' stays positive: a stationary point but not a turning point (a point of inflexion).

Full method - how and why

Reading a factorised derivative. A squared factor does **not** change the sign, so it gives a stationary point that is not a turning point.

- 1 The stationary points are the roots of f' : $x = -1$ and $x = 2$.
why: $f'(x) = 0$ at both.
- 2 Near $x = -1$: $(x - 2)^2 > 0$ always, so the sign of f' is the sign of $(x + 1)$ - negative then positive.
why: A sign change from $-$ to $+$ means the function falls then rises: a local minimum.
- 3 Near $x = 2$: $(x + 1) > 0$ there, and $(x - 2)^2 \geq 0$, so $f' \geq 0$ on both sides - no sign change.
why: A repeated (even-power) factor touches zero without crossing, so the gradient does not change sign.
- 4 Conclusion: local minimum at $x = -1$; stationary but **not** a turning point at $x = 2$.
why: That is a stationary point of inflexion.

Where it goes wrong

- Assuming every root of f' is a turning point.
- Using $f''(2) = 0$ to conclude 'maximum' or 'minimum' - a zero second derivative is inconclusive.

Takeaway Odd-power factors of f' change sign (turning points); even-power factors do not (inflexions).

Question 6 DIF-15 · Differentiation · spec MM6.3

The volume of a cuboid with a square base of side x is $V = 4x^2 - x^3$ for $0 < x < 4$. Find the maximum volume.

Answer: E

$$V' = 8x - 3x^2 = x(8 - 3x) = 0 \Rightarrow x = \frac{8}{3}. V = \frac{64}{9} \cdot 4 - \frac{512}{27} = \frac{768 - 512}{27} = \frac{256}{27}.$$

Full method - how and why

Maximise a cubic on an interval. Differentiate, find the interior stationary point, and evaluate.

- 1 Differentiate: $V' = 8x - 3x^2 = x(8 - 3x)$.
why: Factorise to find the roots easily.
- 2 Roots at $x = 0$ and $x = \frac{8}{3}$. Only $x = \frac{8}{3}$ is inside $0 < x < 4$ with V' changing from $+$ to $-$ there.
why: $x = 0$ is an endpoint (and gives zero volume), so the maximum is at $\frac{8}{3}$.
- 3 Evaluate: $V = 4\left(\frac{8}{3}\right)^2 - \left(\frac{8}{3}\right)^3 = \frac{256}{9} - \frac{512}{27}$.
why: $\left(\frac{8}{3}\right)^2 = \frac{64}{9}$, so $4 \times \frac{64}{9} = \frac{256}{9}$; and $\left(\frac{8}{3}\right)^3 = \frac{512}{27}$.
- 4 Common denominator 27: $\frac{768 - 512}{27} = \frac{256}{27}$.
why: $\frac{256}{9} = \frac{768}{27}$.

Where it goes wrong

- Arithmetic slips with the fractions - $\frac{128}{27}$ and $\frac{64}{27}$ are both offered.
- Taking $x = 0$ as the maximiser.

Takeaway Optimise on an interval: check interior stationary points **and** whether they are maxima; keep fractions exact.

Question 7 DIF-16 · Differentiation · spec MM6.3

The tangent to $y = x^3$ at the point (a, a^3) , where $a \neq 0$, meets the curve again at the point with x -coordinate

Answer: B

Tangent: $y = 3a^2x - 2a^3$. Solve $x^3 - 3a^2x + 2a^3 = 0$; $x = a$ is a double root, so $(x - a)^2(x + 2a) = 0$: the other intersection is $x = -2a$.

Full method - how and why

A tangent meeting the curve again. The tangency shows up as a repeated root, which lets you factorise the cubic without solving it.

1 Find the tangent at (a, a^3) : $y' = 3x^2$, so the gradient is $3a^2$ and the line is $y = 3a^2(x - a) + a^3 = 3a^2x - 2a^3$.

why: Point-gradient form, then simplify.

2 Set the tangent equal to the curve: $x^3 = 3a^2x - 2a^3$, i.e. $x^3 - 3a^2x + 2a^3 = 0$.

why: Intersections satisfy both equations.

3 $x = a$ is a **repeated** root (that is what tangency means), so factor out $(x - a)^2$: $x^3 - 3a^2x + 2a^3 = (x - a)^2(x + 2a)$

why: Check by comparing the constant term: $(-a)^2(2a) = 2a^3$ ✓.

4 The remaining root is $x = -2a$.

why: That is where the tangent cuts the curve again.

Where it goes wrong

- Forgetting that tangency gives a **double** root and trying to factor out only one $(x - a)$.
- Sign error giving $x = 2a$.

Takeaway A tangent meets the curve in a repeated root; factor $(x - a)^2$ out and read off the third root.

Question 8 DIF-19 · Differentiation · spec MM6.1

Which of the following statements is/are true for every function f that has a derivative at every real number?

- I If $f'(a) = 0$ then f has a maximum or a minimum at $x = a$.
- II If f has a local minimum at $x = a$ then $f'(a) = 0$.
- III If $f'(x) > 0$ for all x , then $f(3) > f(2)$.

Answer: G

I is false: $f(x) = x^3$ has $f'(0) = 0$ but no turning point. II is true (a stationary point is required at an interior local extremum of a differentiable function). III is true: a strictly increasing function has $f(3) > f(2)$.

Full method - how and why

Three general claims about differentiable functions. The first is the classic false one; the others are standard facts.

- 1 I: test $f(x) = x^3$ at $a = 0$: $f'(0) = 0$ but the function is increasing throughout, so there is no maximum or minimum. **False.**
why: A stationary point can be a point of inflexion.
- 2 II: at an interior local minimum of a differentiable function the derivative must vanish (otherwise you could move downhill in one direction). **True.**
why: This is Fermat's stationary-point theorem, and is why we solve $f' = 0$ in the first place.
- 3 III: if $f' > 0$ everywhere then f is strictly increasing, so $f(3) > f(2)$. **True.**
why: Positive gradient throughout means the value rises as x rises.
- 4 II and III only.
why: Match to the options.

Where it goes wrong

- Accepting I because 'we find maxima and minima by setting $f' = 0$ ' - that finds candidates, not guarantees.
- Doubting II by thinking of endpoints, which the word 'every real number' excludes.

Takeaway $f'(a) = 0$ is necessary but not sufficient for a turning point; inflexions are the counterexample.

Question 9 X2-02 · Differentiation · spec MM6.3

Let f be a polynomial. P : ' $f'(x) > 0$ for all real x '. Q : ' $f(x) = 0$ has exactly one real root'. Which one of the following is true?

Answer: C

If $f' > 0$ everywhere, f is strictly increasing and (being an odd-degree polynomial) crosses the axis exactly once: P is sufficient. But $f(x) = x^3$ has exactly one root and $f'(0) = 0$, so P is not necessary.

Full method - how and why

A necessary/sufficient question about polynomials. Test each direction, and look for a boundary case to break the converse.

- 1 Sufficiency: if $f'(x) > 0$ for all x , then f is strictly increasing, so it takes each value at most once - at most one root.
why: A strictly increasing function cannot return to a value it has already passed.
- 2 Moreover such an f must have odd degree (an even-degree polynomial turns), so it goes from $-\infty$ to $+\infty$ and crosses the axis. Exactly one root: $P \Rightarrow Q$. **Sufficient.**
why: Both halves are needed: 'at most one' plus 'at least one'.
- 3 Necessity: find an f with exactly one root but f' not always positive. Take $f(x) = x^3$: one root at 0, but $f'(0) = 0$.
why: A stationary point of inflexion satisfies Q without satisfying P .
- 4 So P is sufficient but not necessary.
why: Match to the list.

Where it goes wrong

- Claiming P is necessary because 'one root suggests increasing' - x^3 and also x^2 -type shifts break that.
- Forgetting that $f' > 0$ forces odd degree, so 'at least one root' still needs justifying.

Takeaway Strictly increasing $\Rightarrow$ at most one root; but a single root does not force a strictly positive derivative (x^3).

Question 10 X2-16 · Differentiation · spec MM6.3

The function f has $f'(x) = (x-1)(x-3)^2$. Which of the following statements is/are true?

- I f has a local minimum at $x = 1$.
- II f has a local maximum at $x = 3$.
- III f is increasing for all $x > 1$.

Answer: F

f' changes from negative to positive at $x = 1$ (minimum). At $x = 3$ the squared factor keeps $f' \geq 0$ on both sides, so no turning point: II is false, and $f' \geq 0$ for $x > 1$ (zero only at 3) makes III true.

Full method - how and why

Classify stationary points by whether f' **changes sign** - a squared factor does not change sign.

- 1 $f'(x) = (x-1)(x-3)^2$, so $f' = 0$ at $x = 1$ and $x = 3$.

why: Stationary points are the roots of f' .

- 2 Sign of f' : $(x-3)^2 \geq 0$ always, so the sign is that of $(x-1)$. Negative for $x < 1$, positive for $x > 1$ (zero at $x = 3$).

why: An even-power factor contributes no sign change.

- 3 I: f' goes negative $\rightarrow$ positive at $x = 1$, so a local **minimum. True.**

why: Falling then rising.

- 4 II: at $x = 3$, f' is positive on both sides, so no turning point - a stationary point of inflexion. **False.**

why: The squared factor is precisely why there is no maximum here.

- 5 III: for $x > 1$, $f' \geq 0$ with equality only at the isolated point $x = 3$, so f is increasing. **True.**

why: An isolated zero of the derivative does not interrupt increase.

- 6 So I and III.

why: Match to the list.

Where it goes wrong

- Assuming every root of f' is a turning point and alternating min/max - the repeated root breaks the alternation.
- Rejecting III because $f'(3) = 0$; increase needs $f' \geq 0$ with no interval of equality.

Takeaway Odd-power factors of f' change sign (turning point); even-power factors do not (stationary inflexion).